Multivariable calculus · Differentiation in two variables

Graphs and diagrams

Every graph and diagram in scope, redrawn as clean vector figures.

Partial derivatives, changes along a path, and a circle defined implicitly · compiled 17 October 2026

S1 Lesson: partial derivatives

S2 Lesson: total derivative

S3 Lesson: implicit curves

Contents

1. Figure G1.1 · A coordinate section and its tangent
2. Figure G2.1 · A changing path and its local rate
3. Figure G3.1 · A circle branch and its tangent

Illustrative graphs computed from the stated equations, with dimensionless axes. Dashed lines denote tangents.

1 Partial derivatives and directional slopes

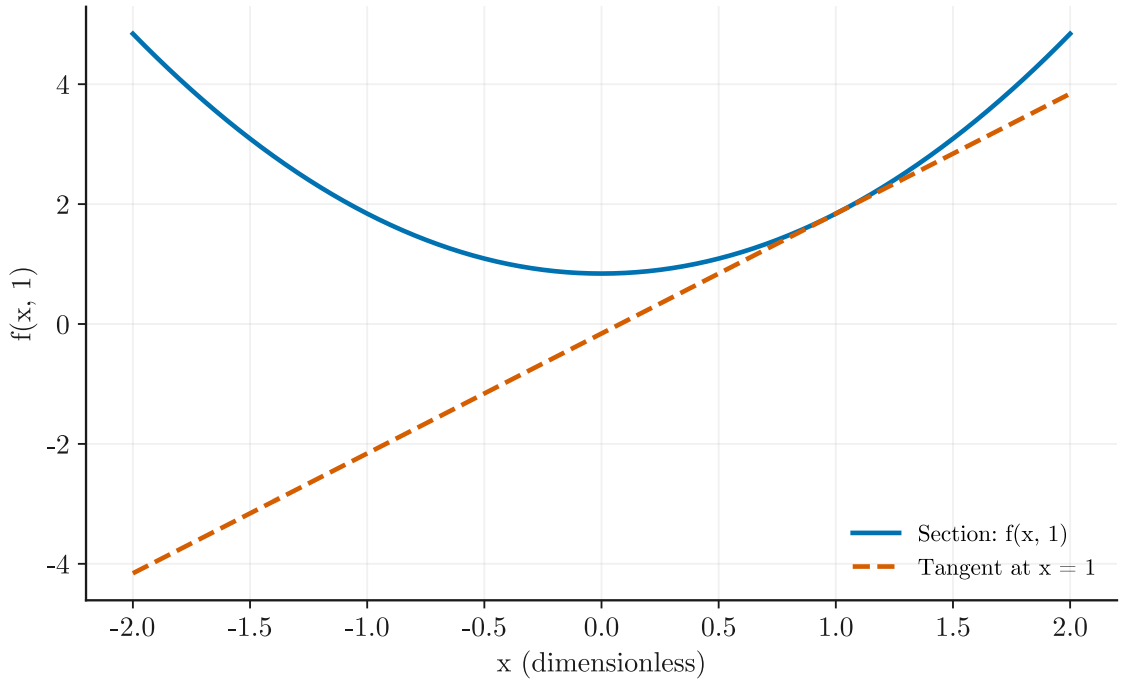


Figure G1.1. A coordinate section and its tangent.

S1

How to read it. Hold y=1 throughout the plot. The derivative with respect to x at x=1 is the slope of the dashed line.

2 Total change and the chain rule

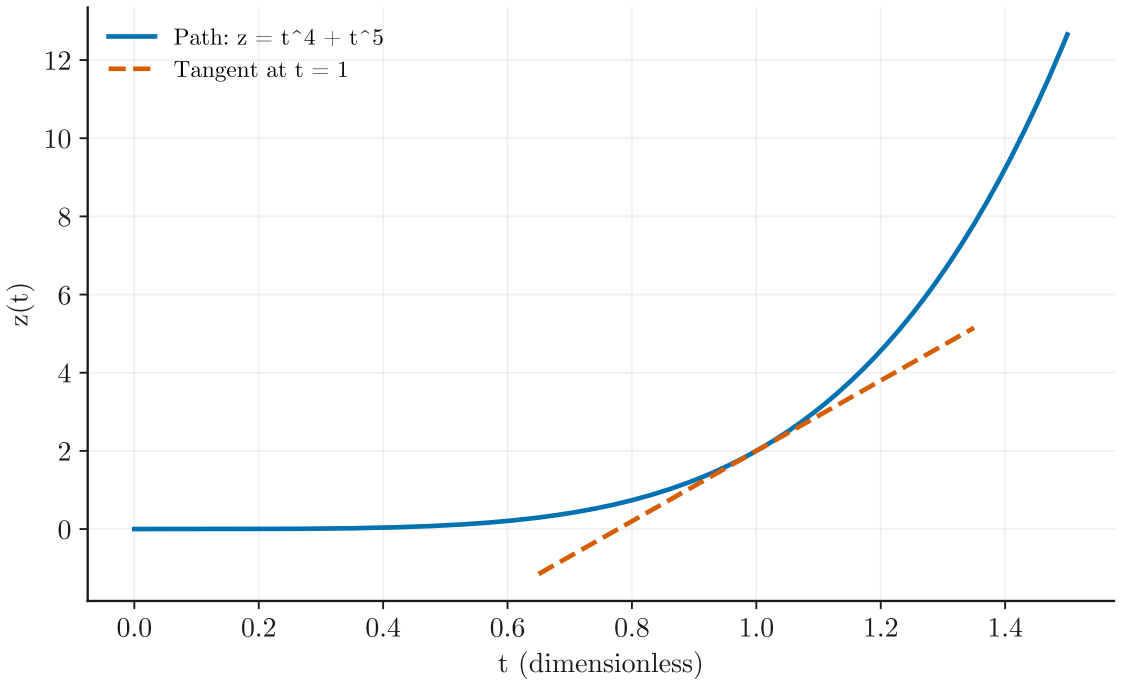


Figure G2.1. A changing path and its local rate.

S2

How to read it. The slope 9 belongs to change along the specified path, with both original inputs varying.

3 Implicit curves and valid tangent formulas

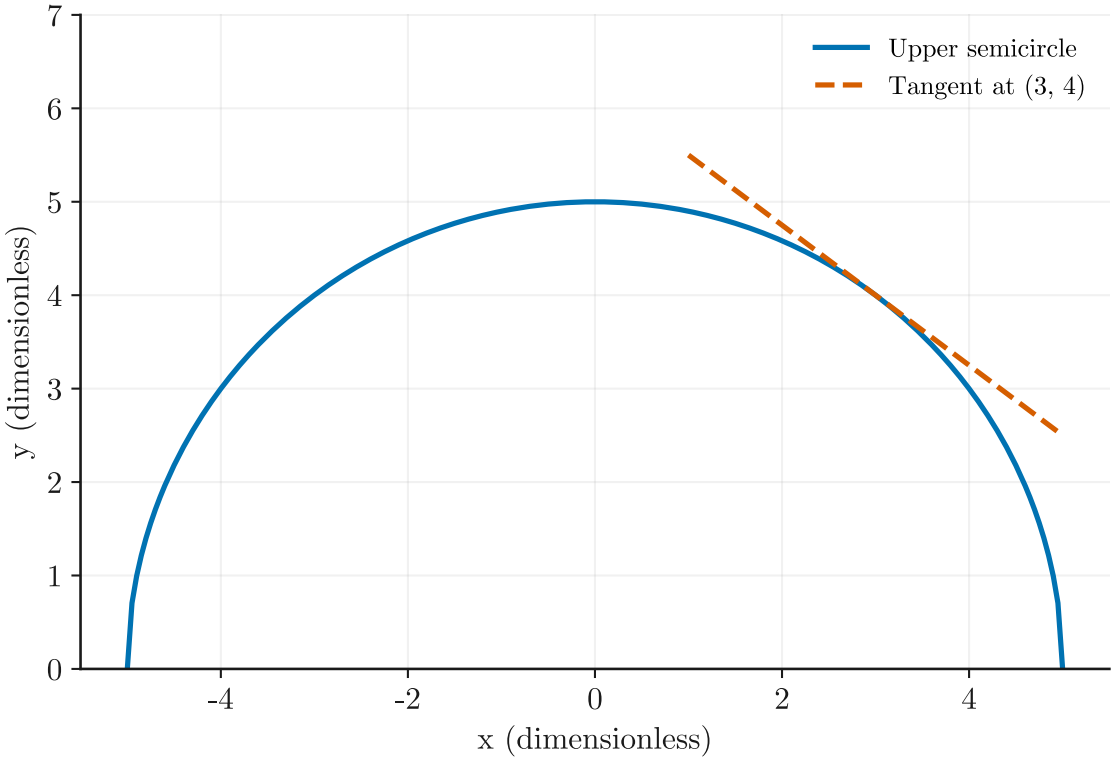


Figure G3.1. A circle branch and its tangent.

S3

How to read it. The tangent is perpendicular to the radius at the contact point. Its downward slope agrees with the implicit derivative.